

Tameem Tantawy

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EDUCATION

University of South Florida

Tampa, FL

Bachelors of science in Computer Science - GPA: 3.86/4.0

Expected Graduation: May 2027

Relevant Courses: Calculus I, Programming Concepts (Python), Program Design (C), Computer Organization, Data Structures, Algorithms (Princeton University), Discrete Structures, Linear Algebra

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript, Go, SQL, HTML, CSS

Frameworks/Libraries: React.js, React Native, Node.js, Express.js, Django, FastAPI, PostgreSQL, MongoDB, SQLAlchemy, NEAT-python, Pygame, Selenium, BeautifulSoup

Developer Tools: Git, Postman, Docker, Amazon Web Services, Heroku

EXPERIENCE

Undergraduate Research Assistant

September 2024 – Present

SHEILD Lab at USF

Tampa, FL

- Developed an app for the Mac Parkman Foundation, aiming to reach over **5,000 individuals** and raise awareness about concussion prevention through engaging educational outreach, featuring live 3D animations, interactive graphs, and real-time metrics.
- Built a scalable backend with Django, leveraging **WebSockets** to support up to **1,000** data points per second received from external sensors, ensuring seamless data flow and minimizing any delay to the frontend.
- Planned software development life cycles using Sprint methodologies, ensuring timely delivery of features through CI/CD pipelines

Co-Founder & Lead Developer

May 2024 – Present

Jamaa

Tampa, FL

- Connected users with their local Muslim community by leading a team to co-develop a cross-platform mobile app, resulting in increased community engagement and participation.
- Designed the frontend using React Native and NativeWind and created user interfaces for the explore page, profile pages, and other components using custom hooks and React's Context API, resulting in page load times decreasing from **200ms to 65ms**.
- Utilized Django's REST Framework to develop **25+** RESTful APIs for different types of users and optimized data storage using table partitioning and indexing in a PostgreSQL database, achieving a **53%** reduction in query response time.
- Secured interest and commitment for testing from **30+** mosques in the Tampa Bay area by successfully pitching the app.

Software Engineering Volunteer

May 2024 – August 2024

Shawarma Bono

Remote

- Increased digital engagement by **25%** with a **MERN**-based full-stack website for a local shawarma restaurant.
- Decreased average page load duration by **40%**, leveraging the capabilities of **React.js** along with **Context API** to facilitate efficient user authentication processes, resulting in improved customer retention rates.
- Improved API response time by **35%** through a **Node.js/Express.js** backend and **MongoDB** database, tested with **50+** Postman cases and secured data with **JWT**.
- Collaborated with the restaurant owner to identify requirements and create a tailored website, enhancing total orders and increasing revenue by **\$24,000**

PROJECTS

Lugha - ShellHacks Project | *FastAPI, PostgreSQL, SQLAlchemy, OpenAI, Groq, Redis* | [Github](#)

- Spearheaded the development of a language learning app with **300+** interactive lessons and quizzes to provide users with structured learning, improving language retention and foundational skills.
- Architected a ranking system using a max-heap, priority queue, and Redis, to match users for voice calls based on skill.
- Engineered a backend system with **FastAPI** and **PostgreSQL**, integrating **OpenAI's Whisper** model for transcription, **Meta's Llama 3.1** model for text generation, and **OpenAI's TTS** model for audio responses in **5+** languages.

HalalMate | *React, FastAPI, PostgreSQL, SQLAlchemy, Selenium, BeautifulSoup, Heroku, AWS* | [Github](#)

- Developed a web scraper using **Selenium** and **BeautifulSoup** to scrape and filter food data from the USF dining halls.
- Developed a **FastAPI** backend integrated with **PostgreSQL**, hosted on **Heroku** and **AWS RDS**, enabling the website to automatically scrape data every **3 hours** for reliable and scalable data storage
- Achieved **99.5%** accuracy in categorizing food data by designing a **React.js** frontend, hosted on **AWS S3**.

Cliff Diver AI | *Python, PyGame, NEAT-python algorithm* | [Github](#)

- Built an interactive cliff diving game that uses **PyGame** and the **NEAT** algorithm to dynamically train an agent through evolving **Neural Networks**, enabling it to master game mechanics.
- Achieved maximum **fitness** for the AI agent after an average of **80** generations, scoring a **97%** dive accuracy.

LEADERSHIP

ColorStack USF

Tech Lead

- Coordinate career-focused workshops in collaboration with industry experts, fostering networking opportunities.
- Hosted a backend development workshop, helping **20+** members to gain hands-on experience and improve their technical proficiency.
- Recruited **50+** underrepresented students in STEM and promoted the ColorStack mission, increasing chapter visibility and engagement.